

These relations form the $\mathcal{N} = 2$ SUSY algebra. It describes SUSY quantum mechanics [38, 6]. We can also view it as generators of a $\mathbb{Z}_2$ -grading of a Hilbert space.

Using these generators it is easy to see that

$$S = q \text{ or } S = q^\dagger$$

realizes nilpotent S 's. On the other hand when

$$S = b \text{ or } S = f,$$

we obtain projector S 's. These operators help construct the constant Y_{ij} operator $S_i S_j$. They can satisfy either of the three conditions set in (3.9). The corresponding Hamiltonians are either given by (3.4) or (3.8).

The operator $Y_{ij} = S_i T_j$ is Baxterized when either one of $ST = 0$ or $TS = 0$ is satisfied. This is easily realized using the SUSY generators⁴ as

$$ST = 0 \left\{ \begin{array}{l} S = q, T = b \\ S = q^\dagger, T = f \\ S = b, T = q^\dagger \\ S = f, T = q. \\ S = b, T = f, \end{array} \right. \quad (3.15)$$

The $TS = 0$ yields the same choices when S and T are interchanged in the above equation. The Hamiltonians in this case are of the form (3.8).

All of the above SUSY realizations are algebraic or independent of the size of the representation of the SUSY algebra. For the $\mathbb{C}^2$ representation the SUSY generators take the form

$$\begin{aligned} q &= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} ; q^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \\ b = qq^\dagger &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} ; f = q^\dagger q = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \quad (3.16)$$

It is clear that these matrices perform a $\mathbb{Z}_2$ -grading of the Hilbert space $\mathbb{C}^2$, with b and f projecting to the ‘bosonic’ and ‘fermionic’ parts of $\mathbb{C}^2$. It is also important to note that the $\mathbb{C}^2$ representation of SUSY (3.16) span the space of operators acting on $\mathbb{C}^2$,

⁴For solutions of the generalized Yang-Baxter equation using similar realizations see [31].